

# No. of Functional Features Included in the Solution

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 2 July 2026                                                  |
| Team ID       | XXX                                                          |
| Project Name  | OptiCrop – Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                      |

## Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S. No | Feature Name          | Feature Description                                   | Module / Component | Status | Marks Contribution |
|-------|-----------------------|-------------------------------------------------------|--------------------|--------|--------------------|
| 1     | Crop Recommendation   | Predicts a suitable crop from soil and climate inputs | Prediction Engine  | Done   | High               |
| 2     | NPK Input             | Accepts nitrogen, phosphorus and potassium values     | Input Module       | Done   | High               |
| 3     | Climate Input         | Accepts temperature, humidity and rainfall values     | Input Module       | Done   | High               |
| 4     | Soil pH Input         | Accepts soil pH value for prediction                  | Input Module       | Done   | High               |
| 5     | Input Validation      | Validates agricultural parameter values               | Validation Module  | Done   | High               |
| 6     | User Interface        | Interactive responsive web interface                  | Frontend           | Done   | Medium             |
| 7     | ML Model Integration  | Connects the trained crop recommendation model        | ML Module          | Done   | High               |
| 8     | Prediction Confidence | Displays crop prediction confidence                   | Result Module      | Done   | Medium             |

## Feature Summary:

| Metric                             | Count / Value                                                   |
|------------------------------------|-----------------------------------------------------------------|
| Total Features Planned             | 8                                                               |
| Total Features Implemented         | 8                                                               |
| Core / Must-Have Features          | Crop Recommendation, NPK Input, Climate Input, Input Validation |
| Additional / Nice-to-Have Features | Prediction Confidence, Responsive UI                            |
| Features Tested & Verified         | 8 Features                                                      |

## Feature Category Breakdown:

| S. No | Category             | Features in Category | Example Features                      |
|-------|----------------------|----------------------|---------------------------------------|
| 1     | User Interface (UI)  | 2                    | Input Form, Prediction Result         |
| 2     | Backend / Logic      | 3                    | Validation, Prediction, Model Loading |
| 3     | Data / Model Storage | 1                    | Crop Dataset and Trained Model        |
| 4     | ML Integration       | 1                    | scikit-learn Crop Model               |

|   |                       |   |                           |
|---|-----------------------|---|---------------------------|
| 5 | Testing / Reliability | 1 | Input and Browser Testing |
|---|-----------------------|---|---------------------------|